

Cnidarians

PHYLUM Cnidaria

CLASSES 6

ORDERS 24

FAMILIES 300

SPECIES About 11,000

This group of simple aquatic animals includes sea anemones, corals, jellyfish, and hydroids.

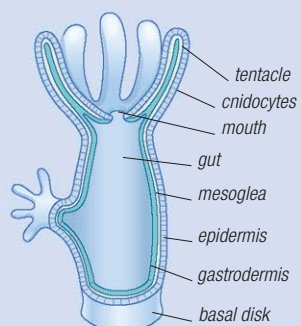
The radially symmetrical body is essentially a tube that is closed at one end and open at the other.

This tube is either flattened into a bell shape (a medusa) or elongated with the closed end

attached to a hard surface (a polyp). In general, medusae swim freely, while polyps live on the seabed. All cnidarians have tentacles around their mouth that contain stinging cells, known as cnidocytes. Most species are marine, although a small number are found in freshwater.

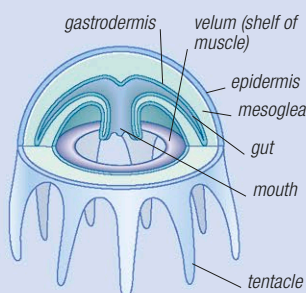
Anatomy

Some cnidarians (such as sea anemones and corals) exist only as polyps. Others (hydroids and some jellyfish) exist as both polyps and medusae at different stages of their life cycle. In both forms, the wall of the body tube consists of just 2 layers of cells: an outer epidermis and an inner gastrodermis, separated by a jellylike matrix (the mesoglea). The central cavity of the tube acts as a gut, and the single opening is used both to take in food and to expel waste. Around the mouth, the body wall is drawn out into one or more whorls of tentacles. These contain the stinging cells, cnidocytes, which are unique to cnidarians. Cnidocytes inject venom into prey and hold it while the tentacles draw it into the mouth.



POLYP

The mouth of a polyp faces outward. At the opposite end of the body is a basal disk, which is often attached to a solid surface. Many polyps, such as this hydra, do not have a well-developed mesoglea. Polyps may occur either singly or in colonies.

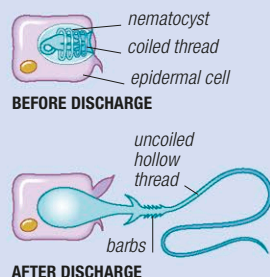


MEDUSA

The body of a medusa looks like a bell, with the mouth at the center of the lower surface and the tentacles at the edges. The mesoglea is thicker than it is in most polyps. Some medusae have a shelf of tissue that is rich in muscle cells and used in locomotion.

CNIDOCYTES

Each cnidocyte contains a bulblike structure, called a nematocyst, which houses a coiled, barbed thread. When triggered by touch or chemicals, the thread explodes outward and pierces the victim's skin. The tentacles are then used to pull the prey in.



Locomotion

Cnidarians can change shape by coordinating the contraction of some of their muscle cells. In polyps, the water inside the body cavity can act in a similar way to a skeleton, providing a resistant structure against which the cells are contracted. A polyp can lengthen, shorten, or bend its body—for example, to reach toward potential prey or to retreat from predators. By alternately relaxing and contracting groups of muscle cells, free-living medusae such as the jellyfish (shown right) can move using a weak form of jet propulsion.



bell fully relaxed, ready to push forward

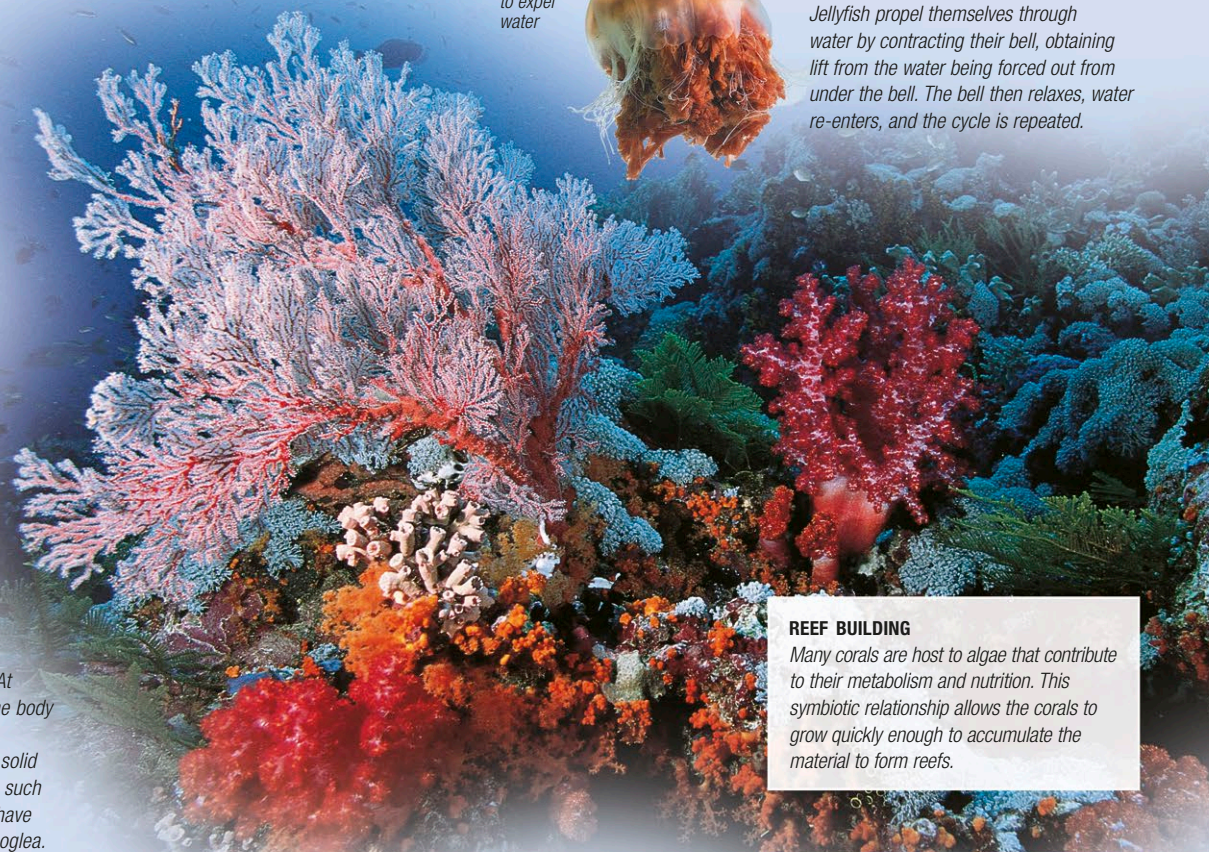


bell begins to relax to take in water

elastic bell contracts to expel water

JELLYFISH MOVING

Jellyfish propel themselves through water by contracting their bell, obtaining lift from the water being forced out from under the bell. The bell then relaxes, water re-enters, and the cycle is repeated.

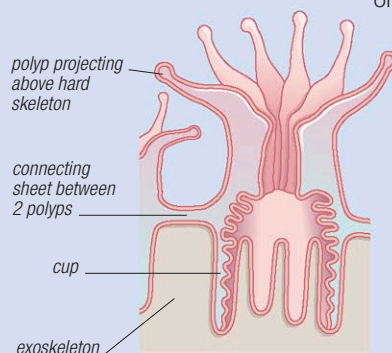


REEF BUILDING

Many corals are host to algae that contribute to their metabolism and nutrition. This symbiotic relationship allows the corals to grow quickly enough to accumulate the material to form reefs.

Colonies

Most cnidarians can reproduce asexually (without mating) by producing buds from their own body. In some cases, the new polyp splits completely; in others, the budding is incomplete so that several individuals remain joined together to form a colony. Colonies can reach great size, as, for example, in reef-forming corals. Some planktonic colonies are highly complex. For example, colonies of siphonophores consist of polyps and medusae integrated into what appears to be an individual organism, with specialized polyps forming the equivalent of organ systems.



CORAL POLYP

The polyps in a coral colony secrete an exoskeleton of hard calcium carbonate, which may accumulate over time to form a reef. The exoskeleton includes structures called cups, into which the polyps can withdraw.

CONSERVATION

Several dozen cnidarian species (mainly corals) are listed as vulnerable and some that form coral reefs are now endangered. Threats include rising sea temperatures—as a result of global warming—discharge of silt in coastal areas, and collection for souvenirs. If the sea temperature rises above a certain level, the corals' symbiotic algae are expelled, and the corals become bleached and often die.

